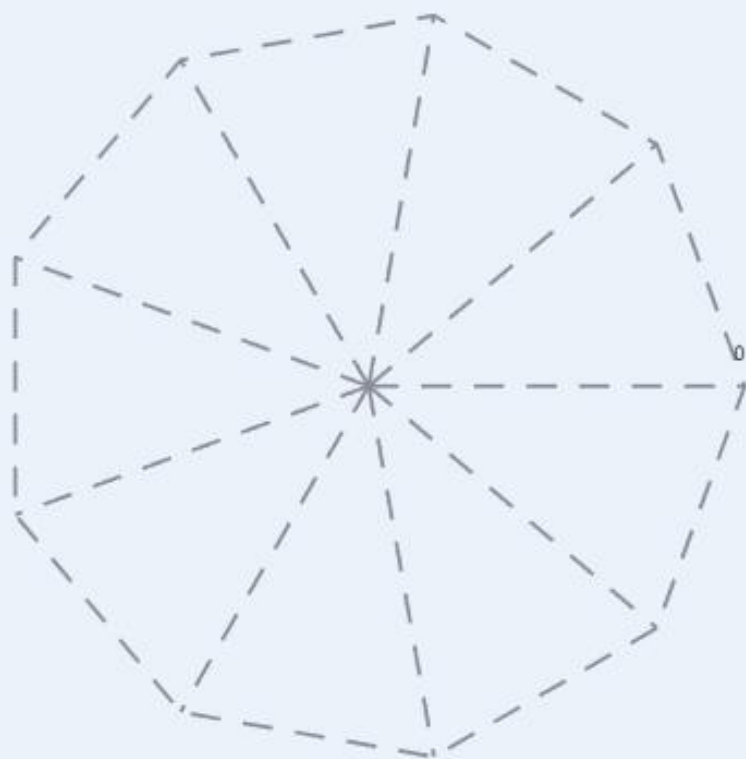




Choose at most three topics that you find most difficult (multiple vote on a single choice is allowed; max 3 votes)



- A 0% Find Cartesian from Parametric Equations
- B 0% Find Tangent Lines to Curve
- C 0% Implicitly defined parametrizations
- D 0% Area enclosed by curve
- E 0% Length of Curves
- F 0% Graph the set of polar coordinates points
- G 0% Polar to Cartesian Equations
- H 0% Cartesian to Polar Equations
- I 0% Nothing is difficult





Consider a curve given by the parametrization $x = f(t)$ and $y = g(t)$. If $\frac{dy}{dt} \Big|_{t=a} \neq 0$ and $\frac{dx}{dt} \Big|_{t=a} = 0$, what can we tell about the tangent line to the curve at the point given by $t = a$?

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\neq 0}{= 0}$$

$$\frac{\text{Nonzero}}{0} = \infty$$

There is no tangent line at that point.

The tangent line is a horizontal line.

Responses Hidden

The tangent line is a vertical line. ✓

$$(x = c)$$

← tangent line

where

$$\frac{dx}{dt} = 0, \frac{dy}{dt} \neq 0$$

The slope of the tangent line is exactly equal to $\frac{dy}{dt} \Big|_{t=a}$.

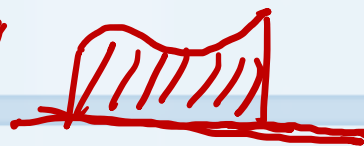
It is not true if $\frac{dy}{dt} = 0$. (out of syllabus)





We want to find the area under $y = x^3$ over $[0, 1]$. Choose the correct expression for the area using the parametrization $x = t^2, y = t^6, 0 \leq t \leq 1$.

Area: ① $\int y \, dx \rightarrow$



② $\int x \, dy \rightarrow$



$$\int_0^1 2t^6 \, dt$$

$$\int_0^1 2t^7 \, dt$$

$$\int_0^1 6t^6 \, dt$$

$$\int_0^1 6t^7 \, dt$$

Responses Hidden

Edit

Show Choices



Show Responses



Lock



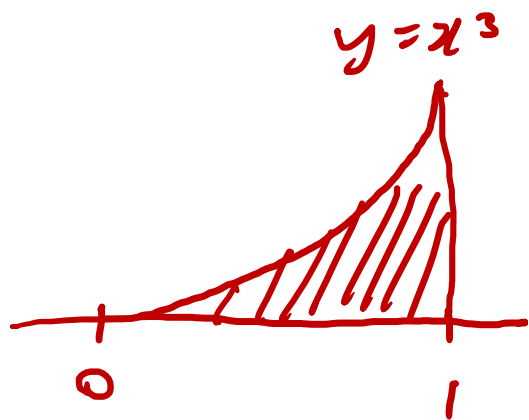
Show Correctness



Ex. 10.2 Q24

(a) $x = t^2$, $y = t^6$

Area: $y = x^3$ over $[0, 1]$



$$\begin{aligned}\Rightarrow \text{Area} &= \int_0^1 y \, dx \\ &= \int_0^1 t^6 \cdot 2t \, dt \\ &= \int_0^1 2t^7 \, dt \\ &= \dots\end{aligned}$$

$$\begin{aligned}x = t^2 &\Rightarrow dx = 2t \, dt \\ \text{When } x = 0, & \quad t = 0 \\ \text{When } x = 1, & \quad t^2 = 1 \Rightarrow t = \pm 1\end{aligned}$$

Choose $t = -1$
we will get
negative
area



Choose the correct graph for $0 \leq \theta \leq \pi/2, 1 \leq |r| \leq 2$

Responses Hidden

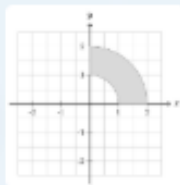
A

0%



B

0%



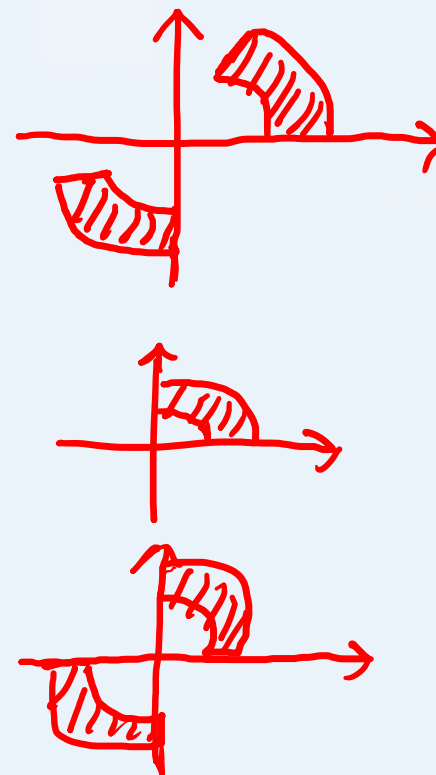
C

0%



D

0%



0 0



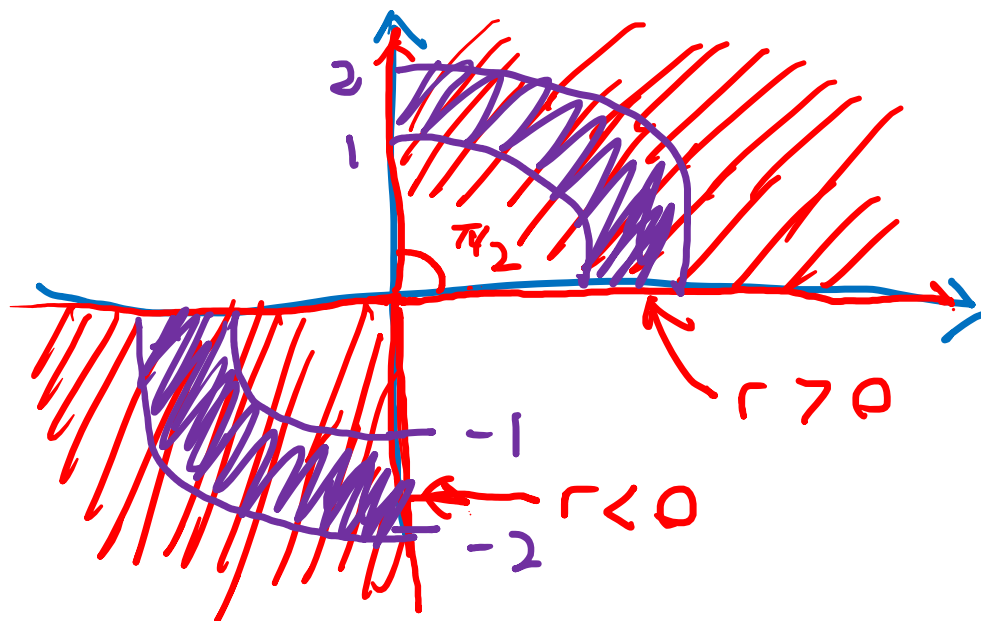
Ex 10.3 Q26

$$0 \leq \theta \leq \pi/2, \quad 1 \leq |r| \leq 2$$



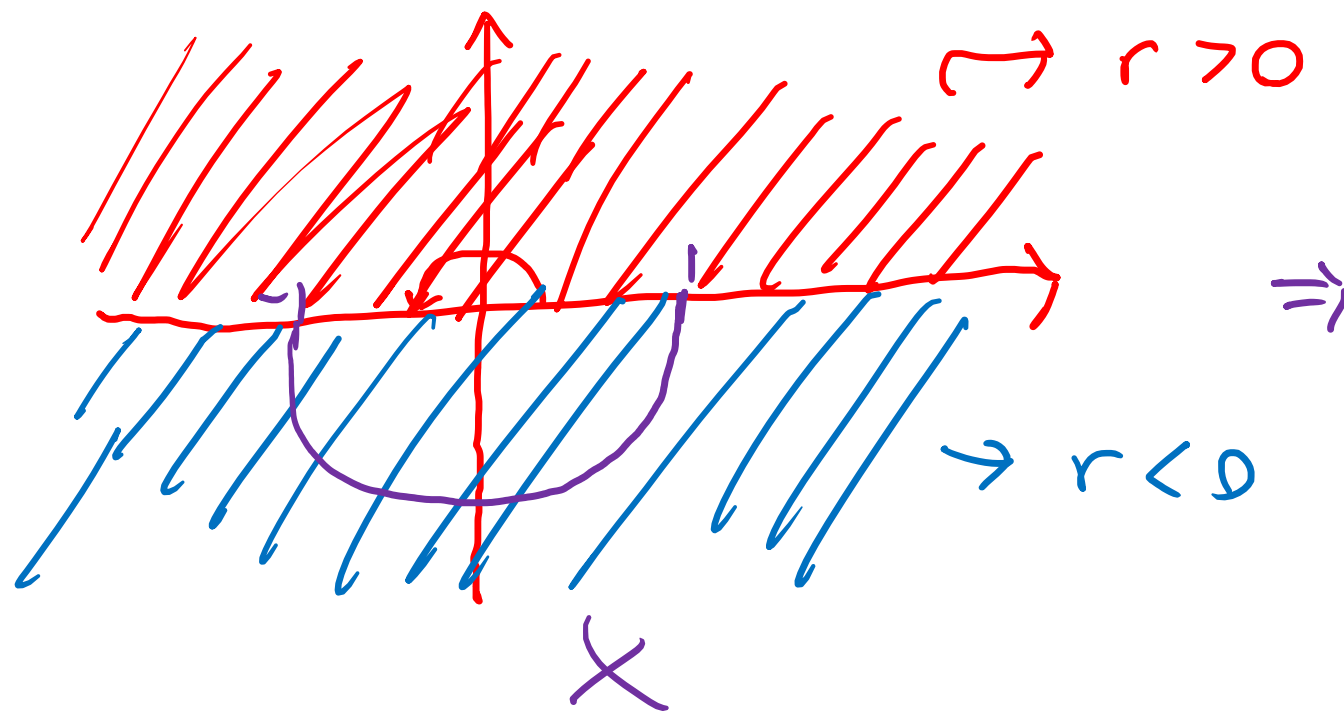
$$1 \leq r \leq 2 \quad \text{or}$$

$$1 \leq -r \leq 2$$
$$-2 \leq r \leq -1$$

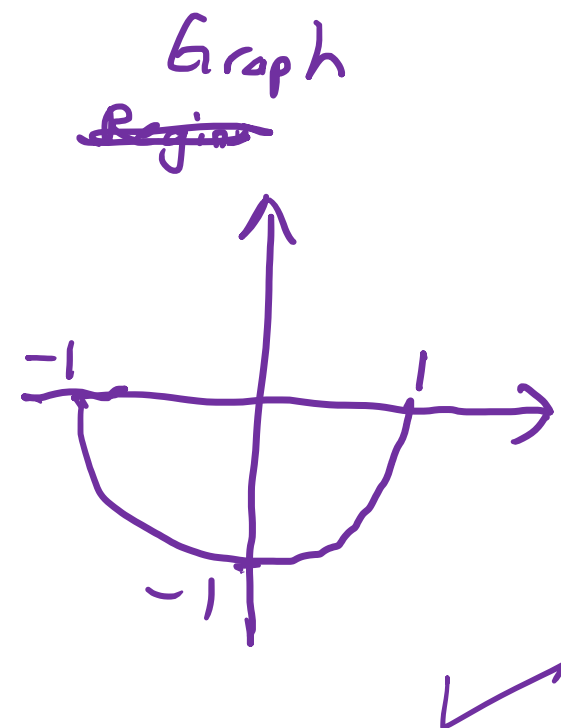


Ex. 10.3 Q22

$0 \leq \theta \leq \pi$, $r = -1$



$\Rightarrow$





When converting cartesian equation $(x - 5)^2 + y^2 = 25$ to polar equation, I write

$$(x^2 - 10x + 25 + y^2 = 25) \Rightarrow x^2 + y^2 = 10x \Rightarrow r^2 = 10r \cos \theta \Rightarrow r = \pm \sqrt{10r \cos \theta}$$

. What do you think about this argument?

$$r^2 = x^2 + y^2, x = r \cos \theta$$

$$10r \cos \theta < 0$$

does not
have r inside

Sounds good to me.

It is flawed.

It is flawed but I cannot reason for that.

It is technically correct but it gives no information on the values of r .

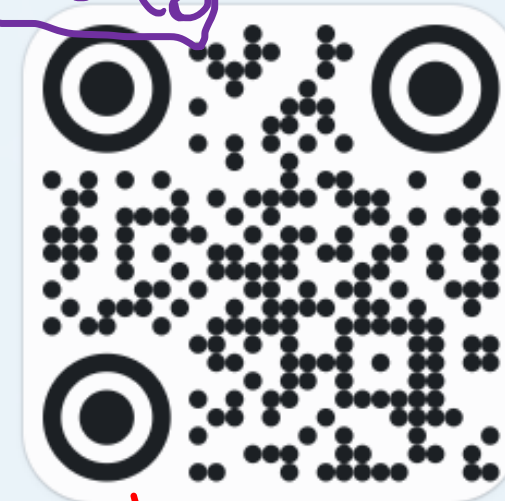
Correct approach:

$$r^2 = 10r \cos \theta$$

$$\Rightarrow r = 10 \cos \theta$$

(divided by r)

($r \neq 0$)



Ex. 10.3 @ 62

$$\underline{x^2 + xy + y^2} = 1 \Rightarrow \underline{(x^2 + y^2) + xy} = 1$$

$$\boxed{x^2 + y^2} = r^2$$

$$x = r \cancel{\sin \theta} \cos \theta$$

$$y = r \sin \theta$$

$$r^2 + (r \cos \theta)(r \sin \theta) = 1$$

$$r^2 (1 + \sin \theta \cos \theta) = 1$$

$$\boxed{\frac{1}{2} \sin(2\theta)} \Rightarrow -1 \leq \sin 2\theta \leq 1$$

$$\Rightarrow -\frac{1}{2} \leq \frac{1}{2} \sin 2\theta \leq \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} \leq 1 + \frac{1}{2} \sin 2\theta \leq \frac{3}{2}$$

for all θ

$$\boxed{r^2 = \frac{1}{1 + \frac{1}{2} \sin 2\theta}}$$